

Francisco Correia de Sousa

|  franciscomcsousa |  franciscomcsousa |  francisco-sousa.pt |
|  francisco.m.c.sousa@gmail.com |  Lisbon, Portugal |

SUMMARY

MSc in Computer Science and Engineering with a focus on **distributed systems**, **network security** and **fault-tolerant protocols**. Experience designing Byzantine-resilient distributed systems and asynchronous protocols.

WORK EXPERIENCE

Teaching Assistant - IST Computer Science Department Nov. 2024 - Feb. 2026

- Taught lab sessions of **Distributed Systems** (Spring 2025) and **Operating Systems** (Fall 2024 & Fall 2025).
- Assisted students in concurrency, synchronization, OS fundamentals and distributed protocols.
- Graded assignments of multiple classes of 20+ students.

Research Intern - INESC-ID June 2024 - Nov. 2026

- Designed Chia-Async, an asynchronous alternative to the Chia blockchain.
- Developed an **Asynchronous DRB** (Distributed Random Beacon) to remove the synchrony assumptions of the protocol.
- Evaluated the protocol's latency, throughput and fault-tolerance under network failures and Byzantine nodes.

PROJECTS

HDS Serenity [Link to Demo](#)

A simplified permissioned blockchain system with high dependability guarantees using IBFT as the consensus protocol.

TheCork [Link to Demo](#)

Full-stack application with special emphasis in encrypted communication, authentication and secure data persistence. Developed using Flask, React, Nginx and MariaDB.

EDUCATION

Instituto Superior Técnico - Computer Science Department Completed Nov. 2025

Computer Science and Engineering **MSc.** GPA: **18/20**

Thesis: "Chia-Async: Towards Asynchronous Proof-of-Space" Grade: **19/20**

Instituto Superior Técnico - Computer Science Department Completed June 2022

Computer Science and Engineering **BSc.** GPA: **15/20**

SKILLS

Technical Areas	Distributed Systems, Network Security, Low-level programming
Programming	C, C++, Python, Java, JS/TS, C#, Go, Rust, SQL
Tools & Systems	Linux, Git, Docker
Language Fluency	Portuguese (Mother Tongue), English (Full Working Proficiency)